EXPERIMENT-5

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Semester: 5
Date: 05/08/2026
Subject Name: Computer Networks
Subject Code: 24CSH-337

AIM:

To understand and implement Dynamic Routing using RIP (Routing Information Protocol) in Cisco Packet Tracer and to study how routers automatically exchange routing information and select paths between different networks.

OBJECTIVES:

- To understand the concept of routing and its importance in computer networks.
- To differentiate between static routing and dynamic routing.
- To understand the working principle of RIP (Routing Information Protocol).
- To configure RIP on multiple Cisco routers using Cisco IOS commands.
- To understand hop count as the routing metric used by RIP.
- To observe how routers automatically learn remote networks.
- To verify connectivity between different networks using the Ping command.
- To examine the routing table and verify dynamically learned routes.

SOFTWARE REQUIREMENTS:

(a) Software:

- Cisco Packet Tracer
- Windows Operating System

(b) Hardware (Virtual):

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- PC-PT
- Cisco 2960-24TT Switch
- Cisco 2911 Router
- Copper Straight-Through Cable
- Copper Cross-Over Cable, if required

THEORY: (i) Routing

Routing is the process of finding the best path for forwarding data packets from a source network to a destination network. When two devices belong to different networks, they cannot communicate directly. A router is required to forward packets between these networks.

If PC1 wants to communicate with PC2 and both PCs belong to different networks, routers determine the appropriate path through which the packet should travel.

Need for Routing:

Requirement	Explanation
Inter-network Communication	Connects different networks
Path Selection	Finds a route to the destination
Scalability	Supports large networks
Internet Communication	Makes communication between networks possible

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(ii) Static Routing

In static routing, routes are manually configured by the network administrator.

If the network connected to R3 fails, R1 does not automatically know about an alternate route. The administrator has to manually modify the routing configuration.

Disadvantages of Static Routing:

Disadvantage	Explanation

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Manual Configuration	Every route must be entered manually
Not Scalable	Difficult to manage in large networks
Time Consuming	Requires more administrative work
No Automatic Updates	Route changes must be configured manually

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(iii) Dynamic Routing

Dynamic routing is a routing technique in which routers automatically learn and update routing information.

Routers exchange routing information with neighbouring routers and automatically learn about networks that are not directly connected.

R1 can automatically learn about networks located behind R2 and R3 through routing protocol updates.

Advantages of Dynamic Routing:

Advantage	Explanation
Automatic Learning	Learns routes automatically
Less Manual Work	No need to configure every route
Fault Tolerance	Can find alternate routes
Scalable	Suitable for larger networks

(iv) Dynamic Routing Protocols

Some commonly used dynamic routing protocols are:

Protocol	Full Form
RIP	Routing Information Protocol
OSPF	Open Shortest Path First

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EIGRP	Enhanced Interior Gateway Routing Protocol
BGP	Border Gateway Protocol

(v) RIP – Routing Information Protocol

RIP stands for Routing Information Protocol. It is a distance-vector routing protocol that automatically exchanges routing information between neighbouring routers.

RIP is one of the oldest and simplest dynamic routing protocols.

RIP was developed to overcome some of the limitations of static routing by providing:

1. Automatic network discovery.
2. Automatic routing table updates.
3. Automatic route selection.

(vi) RIP Metric – Hop Count

RIP uses hop count as its routing metric.

A hop represents one router crossed by a packet while reaching the destination network.

For example:

From R1, the packet crosses R2 and R3 before reaching the destination network.

Therefore, the path has a hop count according to the routers crossed.

Hop Count	**Meaning**
1	One router
2	Two routers
5	Five routers
15	Maximum reachable
16	Unreachable network

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The maximum reachable hop count in RIP is 15. A route with a hop count of 16 is considered unreachable.

(vii) RIP Route Selection

RIP selects a route based on the lowest hop count.

Eg: Path 1: R1 → R2 → Network

Hop Count = 1

Path 2: R1 → R3 → R4 → Network

Hop Count = 2

RIP selects Path 1 because it has fewer hops.

(viii) RIP Routing Table

A routing table may contain entries such as:

Destination Network	**Next Hop**	**Hop Count**
192.168.1.0	Directly Connected	0
192.168.2.0	10.0.0.2	1
192.168.3.0	10.0.0.2	2

(ix) RIP Route Updates

Routers periodically exchange routing information with neighbouring routers.

The default RIP update interval described in the provided document is 30 seconds. During these updates, routers send their routing information to neighbouring routers.

Eg: R1 knows → 192.168.1.0

R2 knows → 192.168.2.0

After RIP updates:

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R1 learns → 192.168.2.0

R2 learns → 192.168.1.0

(x) RIP Convergence

Convergence is the process through which all routers update their routing information and agree on the current network topology.

For example, if an R2 link fails, routers update their routing tables. Once all routers know the new available path, the network is considered converged.

STEPS/IMPLEMENTATION:

Part I – Creating the Network Topology

1. Open Cisco Packet Tracer.
2. Place three Cisco 2911 Routers in the workspace.
3. Place three Cisco 2960-24TT Layer-2 Switches.
4. Place six PC-PT devices.
5. Connect PC0 and PC1 to Switch0 using Copper Straight-Through cables.
6. Connect PC2 and PC3 to Switch1 using Copper Straight-Through cables.
7. Connect PC4 and PC5 to Switch2 using Copper Straight-Through cables.
8. Connect Switch0 to Router3, Switch1 to Router4, and Switch2 to Router5.
9. Connect Router3 to Router4 and Router4 to Router5 to form the interconnected network.

Part II – Assigning IP Addresses

10. Open each PC and select Desktop → IP Configuration.
11. Assign the required IPv4 address and subnet mask to each PC.

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12. Configure the default gateway of each PC according to the router interface connected to its LAN.

IP Address Configuration:

Device	IP Address	Subnet Mask	Default Gateway
PC0	192.168.1.1	255.255.255.0	192.168.1.3
PC1	192.168.1.2	255.255.255.0	192.168.1.3
PC2	192.168.2.1	255.255.255.0	192.168.2.3
PC3	192.168.2.2	255.255.255.0	192.168.2.3
PC4	192.168.3.1	255.255.255.0	192.168.3.3
PC5	192.168.3.2	255.255.255.0	192.168.3.3

Part III- Configuring Router Interfaces Using Config Tab

13. Open **Router3** and select the Config tab.
14. Select the appropriate GigabitEthernet interface.
15. Assign the IP address **192.168.1.3** with subnet mask 255.255.255.0 for the LAN connection.
16. Configure the router interface connected to Router4 with IP address 192.168.4.1**.**
17. Turn Port Status ON for the configured interfaces.
18. Open Router4 → Config**.**
19. Configure the interface connected to Router3 with IP address 192.168.4.2**.**
20. Configure the LAN interface with IP address 192.168.2.3**.**
21. Configure the interface connected to Router5 with IP address 192.168.5.1**.**
22. Turn Port Status ON for the configured interfaces.
23. Open Router5 → Config**.**
24. Configure the interface connected to Router4 with IP address 192.168.5.2**.**

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25. Configure the LAN interface with IP address 192.168.3.3**.**

26. Turn Port Status ON for the configured interfaces.

Part IV – Configuring RIP

27. Open Router3 and go to the Config tab.
28. Select Routing → RIP.
29. Add the directly connected networks:

- 192.168.1.0
- 192.168.4.0

30. Open Router4 → Config → Routing → RIP.
31. Add the directly connected networks:

- 192.168.2.0
- 192.168.4.0
- 192.168.5.0

32. Open Router5 → Config → Routing → RIP.
33. Add the directly connected networks:

- 192.168.3.0
- 192.168.5.0

34. Allow the routers to exchange routing information through RIP.
35. Observe that the routers automatically learn the remote networks through dynamic routing.

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Part V – Verification of Routing

36. Open the CLI of each router only for verification purposes.
37. Use the command:

    show ip route

38. Observe the routing table of each router.
39. Verify that dynamically learned routes are displayed with the R designation for RIP.
40. Use:

    show ip interface brief

to verify that the configured interfaces are operational.

Part VI – Testing Connectivity

41. Open PC0 → Desktop → Command Prompt.
42. Ping PC2 using:

    ping 192.168.2.1

43. Ping PC4 using:

    ping 192.168.3.1

44. Open PC2 and test connectivity with PC0:

    ping 192.168.1.1

45. Open PC4 and test connectivity with PC0:

    ping 192.168.1.1

46. Observe the successful replies, confirming that communication between the different networks has been established through RIP.

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PROCEDURE:

1. Network Topology

- Open Cisco Packet Tracer.
- Place three routers, three switches, and six PCs.
- Connect the PCs to their respective switches.
- Connect each switch to its router.
- Connect the three routers in sequence.

2. IP Configuration

- Open each PC.
- Go to Desktop → IP Configuration.
- Enter the IP address, subnet mask, and default gateway.
- Configure the router interface IP addresses from the Config tab rather than assigning them through CLI.

3. Router Interface Configuration

- Open each router.
- Select Config → Interfaces.
- Select the required GigabitEthernet interface.
- Enter the appropriate IP address and subnet mask.
- Enable the interface using Port Status: On.

4. RIP Configuration

- Open Config → Routing → RIP on each router.

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- Enter the networks directly connected to that router.
- Configure RIP for all three routers.
- Allow the routers to exchange routing information automatically.

5. Routing Table Verification

Use the router CLI to verify the routing table:

```
show ip route
```

6. Connectivity Verification

- Open the Command Prompt of different PCs.
- Use the ping command to test communication between different LANs.
- Successful replies confirm that dynamic routing has been implemented correctly.

INPUT/OUTPUT DETAILS:

(a) Devices Used

- PC-PT × 6
- Cisco 2960-24TT Switch × 3
- Cisco 2911 Router × 3

(b) Networks Used

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24
- 192.168.4.0/24
- 192.168.5.0/24

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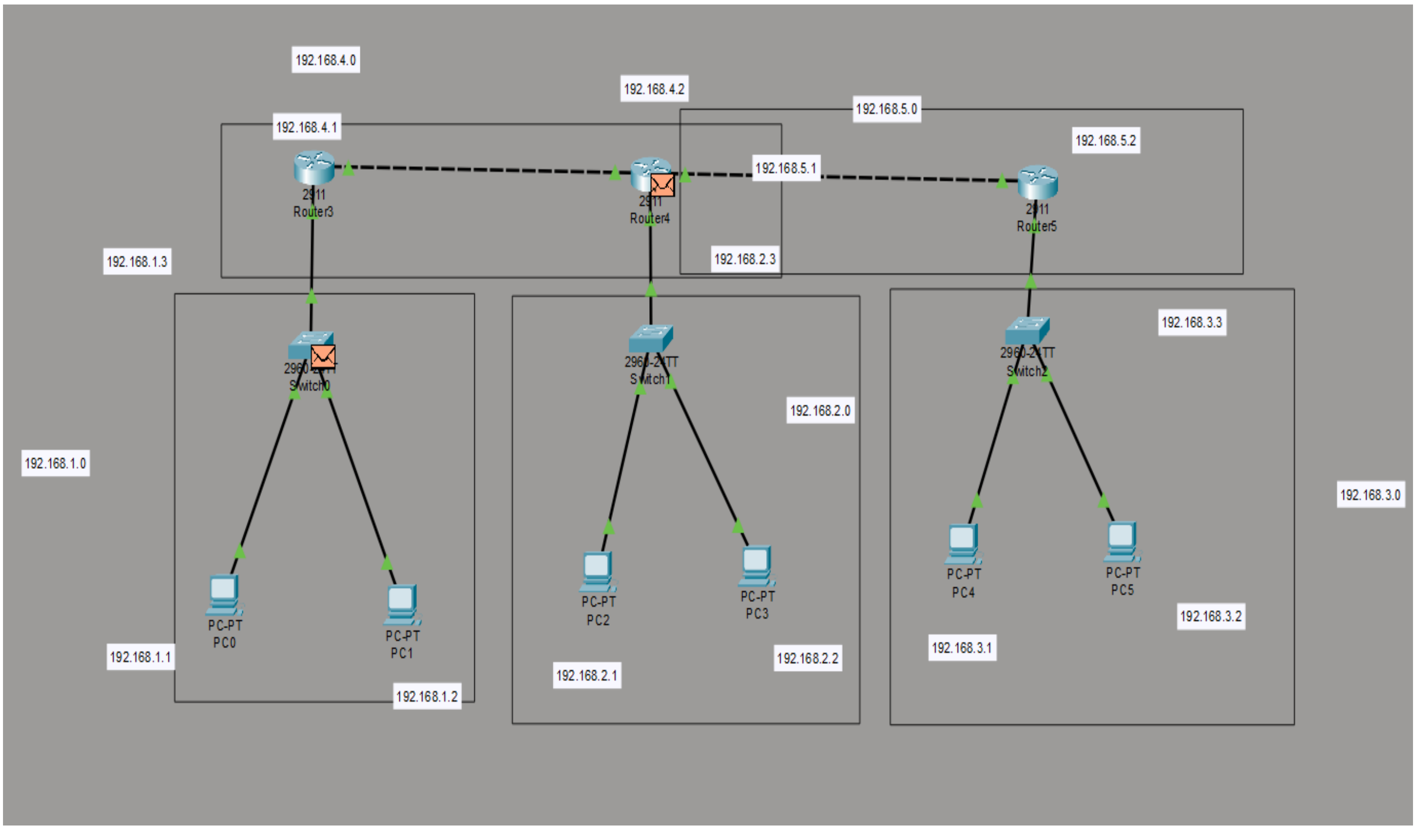
(c) Cisco IOS Commands Used

- enable
- configure terminal
- interface gigabitEthernet
- ip address
- no shutdown
- show ip interface brief
- show ip route
- ping

DEMONSTRATION/OUTPUT:

Part-1: Network Topology



Part-II: IP Address Configuration

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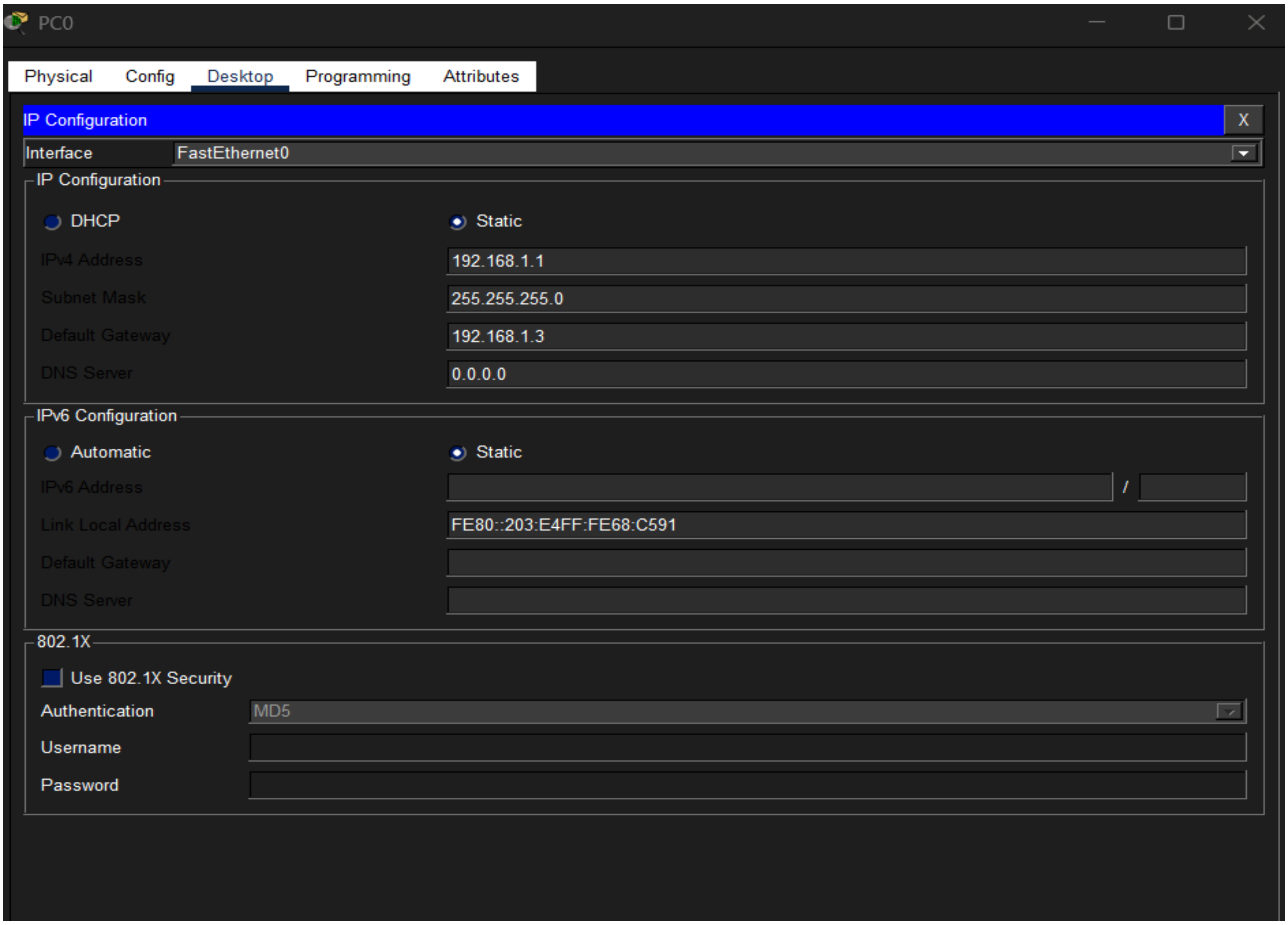

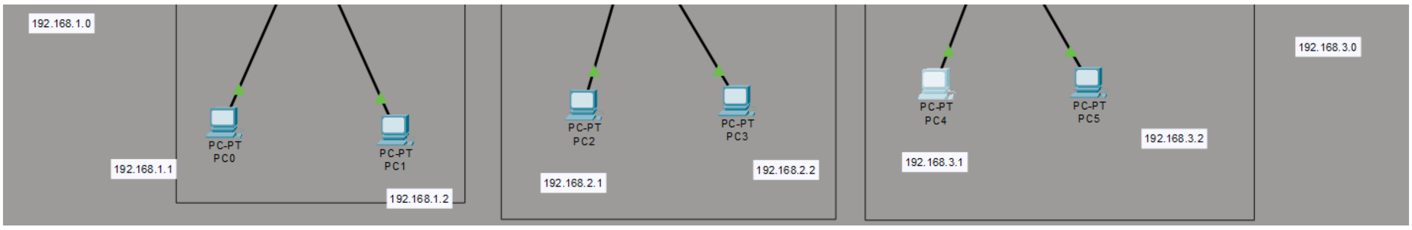

Part III: RIP Configuration

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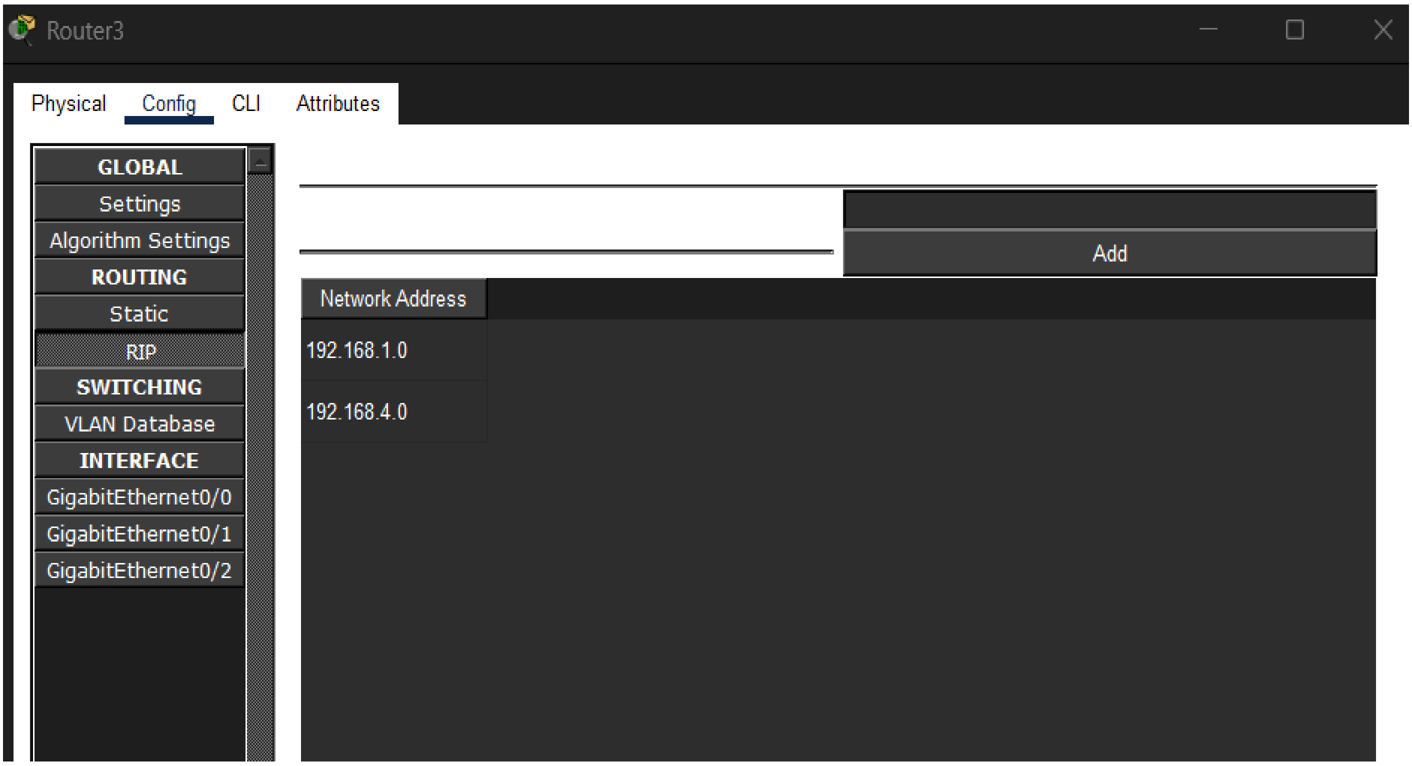


```
Router#
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#router rip
Router(config-router)#
Router(config-router)#end
Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
Router(config)#
Router(config)#router rip
Router(config-router)#
```

Part IV: Routing Table Verification

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```
Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

R    192.168.1.0/24 [120/2] via 192.168.5.1, 00:00:24, GigabitEthernet0/1
R    192.168.2.0/24 [120/1] via 192.168.5.1, 00:00:24, GigabitEthernet0/1
     192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/0
L       192.168.3.3/32 is directly connected, GigabitEthernet0/0
R    192.168.4.0/24 [120/1] via 192.168.5.1, 00:00:24, GigabitEthernet0/1
     192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.5.0/24 is directly connected, GigabitEthernet0/1
L       192.168.5.2/32 is directly connected, GigabitEthernet0/1
```

Part V: Connectivity Test

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time<1ms TTL=126
Reply from 192.168.2.1: bytes=32 time<1ms TTL=126
Reply from 192.168.2.1: bytes=32 time<1ms TTL=126
Reply from 192.168.2.1: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

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RESULT: The experiment was successfully completed in Cisco Packet Tracer. Dynamic routing using RIP (Routing Information Protocol) was implemented on multiple routers. The routers exchanged routing information and automatically learned remote networks.

The routing tables were verified using the show ip route command, and connectivity between different LANs was successfully tested using the ping command.

RIP uses hop count as its routing metric and selects the route with the lowest hop count.

LEARNING OUTCOMES:

After completing this experiment:

- I understood the concept of routing and dynamic routing by learning how routers forward packets between different networks and automatically exchange routing information with neighboring routers.
- I gained practical knowledge of RIP configuration by configuring RIP on multiple Cisco routers and advertising their directly connected networks using Cisco IOS commands.
- I understood the concept of hop count as a routing metric and learned that RIP selects routes based on the number of routers that must be crossed to reach a destination network.
- I learned how routers dynamically update their routing tables by exchanging routing information with neighboring routers, reducing the need for manually configured routes.
- I developed practical skills in network configuration and troubleshooting by configuring IP addresses, router interfaces, RIP, routing tables, and verifying end-to-end connectivity using the Ping command.